Integral Anisotropic Gaussian Kernels:
Formal Definitions

1 Preliminaries

Definition 1.1 (Cone of Positive Semi-Definite Matrices)**.** Let $d \geq 1$. The *cone of symmetric positive semi-definite matrices* is

$$\mathcal{S}_+^d = \left\{ A \in \mathbb{R}^{d\times d} : A = A^\top \text{ and } x^\top A x \geq 0 \ \ \forall x \in \mathbb{R}^d \right\}. \tag{1}$$

The interior $\mathcal{S}_{++}^d$ denotes the *positive definite* matrices (strict inequality for $x \neq 0$).

Definition 1.2 (Anisotropic Gaussian Kernel)**.** For $A \in \mathcal{S}_+^d$, the *anisotropic Gaussian kernel* with precision matrix A is

$$K_A : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}, \quad K_A(x,y) = \exp\left(-\frac{1}{2}(x-y)^\top A (x-y) \right). \tag{2}$$

Definition 1.3 (Borel Measures on $\mathcal{S}_+^d$)**.** Equip $\mathcal{S}_+^d$ with the subspace topology inherited from $\mathbb{R}^{d\times d}$ and let $\mathcal{B}(\mathcal{S}_+^d)$ denote the Borel σ-algebra. Let $\mathcal{M}_+(\mathcal{S}_+^d)$ denote the space of finite non-negative Borel measures on $\mathcal{S}_+^d$, and let $\mathcal{P}(\mathcal{S}_+^d) \subset \mathcal{M}_+(\mathcal{S}_+^d)$ denote the probability measures.

2 The Integral Anisotropic Gaussian Kernel

Definition 2.1 (Integral Anisotropic Gaussian Kernel)**.** Let $\theta \in \mathcal{M}_+(\mathcal{S}_+^d)$ be a finite non-negative Borel measure on the cone of positive semi-definite matrices. The *integral anisotropic Gaussian kernel* induced by θ is the function $K_\theta : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ defined by

$$\boxed{K_\theta(x,y) = \int_{\mathcal{S}_+^d} K_A(x,y)\, d\theta(A) = \int_{\mathcal{S}_+^d} \exp\left(-\frac{1}{2}(x-y)^\top A (x-y) \right) d\theta(A).} \tag{3}$$

We call θ the *mixing measure* or *structure measure* of the kernel.

Remark 2.2. Special cases:

(i) If $\theta = \sum_{j=1}^J a_j \delta_{A_j}$ is a finite weighted sum of Dirac masses, then

$$K_\theta(x,y) = \sum_{j=1}^J a_j\, K_{A_j}(x,y), \tag{4}$$

recovering the finite mixture of anisotropic Gaussian kernels.

(ii) If $\theta \in \mathcal{P}(\mathcal{S}_+^d)$ is a probability measure, K_θ is a *scale-mixture* of anisotropic Gaussians.

3 Well-Definedness and Basic Properties

Proposition 3.1 (Well-Definedness). *For any $\theta \in \mathcal{M}_+(\mathcal{S}_+^d)$, the integral in Definition **??** is well-defined and finite for all $x, y \in \mathbb{R}^d$.*

Proof. For all $A \in \mathcal{S}_+^d$ and $x, y \in \mathbb{R}^d$, we have $(x-y)^\top A(x-y) \geq 0$, hence

$$0 < K_A(x, y) = \exp\left(-\frac{1}{2}(x-y)^\top A(x-y)\right) \leq 1. \tag{5}$$

Therefore,

$$0 < K_\theta(x, y) \leq \int_{\mathcal{S}_+^d} 1 \, d\theta(A) = \theta(\mathcal{S}_+^d) < \infty. \tag{6}$$

□

Proposition 3.2 (Positive Semi-Definiteness). *For any $\theta \in \mathcal{M}_+(\mathcal{S}_+^d)$, the kernel K_θ is positive semi-definite.*

Proof. Each K_A is positive semi-definite. For any $n \in \mathbb{N}$, points $\{x_i\}_{i=1}^n \subset \mathbb{R}^d$, and coefficients $\{c_i\}_{i=1}^n \subset \mathbb{R}$:

$$\sum_{i,j=1}^n c_i c_j K_\theta(x_i, x_j) = \int_{\mathcal{S}_+^d} \underbrace{\sum_{i,j=1}^n c_i c_j K_A(x_i, x_j)}_{\geq 0 \text{ since } K_A \text{ is PSD}} d\theta(A) \geq 0. \tag{7}$$

□

Proposition 3.3 (Stationarity). *The kernel K_θ is stationary: $K_\theta(x, y) = \phi_\theta(x-y)$ where*

$$\phi_\theta(z) = \int_{\mathcal{S}_+^d} \exp\left(-\frac{1}{2} z^\top A z\right) d\theta(A). \tag{8}$$

Proposition 3.4 (Normalization). *$K_\theta(x, x) = \theta(\mathcal{S}_+^d)$ for all $x \in \mathbb{R}^d$.*

4 Fourier Representation

Proposition 4.1 (Spectral Representation). *If θ is supported on $\mathcal{S}_{++}^d$ and satisfies $\int_{\mathcal{S}_{++}^d} |A|^{-1/2} d\theta(A) < \infty$, then*

$$K_\theta(x, y) = \int_{\mathbb{R}^d} e^{i\omega^\top (x-y)} \, p_\theta(\omega) \, d\omega, \tag{9}$$

where the spectral density *is the mixture of Gaussians:*

$$p_\theta(\omega) = \int_{\mathcal{S}_{++}^d} \frac{|A|^{1/2}}{(2\pi)^{d/2}} \exp\left(-\frac{1}{2}\omega^\top A^{-1} \omega\right) d\theta(A). \tag{10}$$

5 The RKHS of K_θ

Theorem 5.1 (RKHS Structure). *The reproducing kernel Hilbert space $\mathcal{H}_{K_\theta}$ of K_θ admits the characterization:*

$$\mathcal{H}_{K_\theta} = \left\{ f = \int_{\mathcal{S}_+^d} f_A \, d\theta(A) : f_A \in \mathcal{H}_{K_A} \text{ for } \theta\text{-a.e. } A, \quad \int_{\mathcal{S}_+^d} \|f_A\|_{\mathcal{H}_{K_A}}^2 d\theta(A) < \infty \right\} \tag{11}$$

with norm given by the continuous infimal convolution*:*

$$\boxed{\|f\|_{\mathcal{H}_{K_\theta}}^2 = \inf\left\{ \int_{\mathcal{S}_+^d} \|f_A\|_{\mathcal{H}_{K_A}}^2 \, d\theta(A) : f = \int_{\mathcal{S}_+^d} f_A \, d\theta(A), \quad f_A \in \mathcal{H}_{K_A} \right\}.} \tag{12}$$

6 The Associated Gaussian Measure

Definition 6.1 (Gaussian Measure Induced by K_θ)**.** The *Gaussian measure* μ_{K_θ} associated with K_θ is the unique Borel probability measure on $L^2(P_X)$ with characteristic functional

$$\hat{\mu}_{K_\theta}(g) = \int_{L^2} e^{i\langle g,f\rangle_{L^2}} d\mu_{K_\theta}(f) = \exp\left(-\frac{1}{2}\langle g, T_{K_\theta} g\rangle_{L^2}\right), \tag{13}$$

where T_{K_θ} is the integral operator with kernel K_θ.

Theorem 6.2 (Decomposition of the Gaussian Process)**.** *If $\theta \in \mathcal{P}(\mathcal{S}_+^d)$ is a probability measure, then*

$$f \sim \mu_{K_\theta} \iff f = \int_{\mathcal{S}_+^d} f_A \, d\theta(A), \quad \text{where } \{f_A\}_{A\in\mathcal{S}_+^d} \text{ are independent with } f_A \sim \mu_{K_A}. \tag{14}$$

More precisely, there exists a probability space supporting a random field $(f_A)_{A\in\mathcal{S}_+^d}$ such that:

(i) For θ-almost every A, $f_A \sim \mathcal{GP}(0, K_A)$.

(ii) The marginals $(f_{A_1}, \ldots, f_{A_n})$ are independent for θ-a.e. distinct $A_1, \ldots, A_n$.

(iii) $f = \int_{\mathcal{S}_+^d} f_A \, d\theta(A)$ satisfies $f \sim \mathcal{GP}(0, K_\theta)$.

7 Common Choices of Mixing Measure θ

Example 7.1 (Finite Mixture)**.**

$$\theta = \sum_{j=1}^{J} a_j \delta_{A_j}, \quad a_j > 0, \quad A_j \in \mathcal{S}_{++}^d. \tag{15}$$

Example 7.2 (Isotropic Scale Mixture)**.** Let ν be a measure on $(0, \infty)$ and set $\theta = \nu \circ \psi^{-1}$ where $\psi(s) = s \cdot I_d$. Then:

$$K_\theta(x, y) = \int_0^\infty \exp\left(-\frac{s}{2}\|x - y\|^2\right) d\nu(s). \tag{16}$$

This recovers Schoenberg's representation for radial positive definite functions.

Example 7.3 (Wishart Mixing Measure)**.** Let $\theta = \mathrm{Wishart}_d(\nu, \Psi)$ with density

$$p(A) = \frac{|A|^{(\nu-d-1)/2}}{2^{\nu d/2}|\Psi|^{\nu/2}\Gamma_d(\nu/2)} \exp\left(-\frac{1}{2}\operatorname{Tr}(\Psi^{-1}A)\right), \quad A \in \mathcal{S}_{++}^d, \tag{17}$$

where $\nu > d - 1$ and Γ_d is the multivariate gamma function.

Example 7.4 (Rank-One (Ridge Function) Measure)**.** Let $\lambda > 0$ and let ρ be a measure on the sphere $\mathbb{S}^{d-1}$. Define θ as the pushforward of ρ under $w \mapsto \lambda \cdot ww^\top$:

$$K_\theta(x, y) = \int_{\mathbb{S}^{d-1}} \exp\left(-\frac{\lambda}{2}(w^\top(x - y))^2\right) d\rho(w). \tag{18}$$

This connects to neural networks and ridge function approximation (Bach, 2017).

Example 7.5 (Spectral Decomposition Measure)**.** Write $A = U\Lambda U^\top$ with $U \in O(d)$ orthogonal and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_d)$. Let

$$d\theta(A) = d\mu(\lambda_1, \ldots, \lambda_d) \, d\nu(U), \tag{19}$$

where μ is a measure on $\mathbb{R}_+^d$ and ν is a measure on the orthogonal group $O(d)$.

If ν is the Haar measure, then K_θ is *rotation-invariant.*

8 Characterization Theorem

Theorem 8.1 (Representation Theorem)**.** *A continuous stationary kernel $K(x, y) = \phi(x - y)$ on $\mathbb{R}^d$ can be represented as*

$$\phi(z) = \int_{\mathcal{S}_+^d} \exp\left(-\frac{1}{2}z^\top Az\right) d\theta(A) \tag{20}$$

for some $\theta \in \mathcal{M}_+(\mathcal{S}_+^d)$ if and only if its spectral measure (via Bochner's theorem) is a mixture of Gaussian measures on $\mathbb{R}^d$.

Equivalently, ϕ is representable if and only if its Fourier transform satisfies

$$\hat{\phi}(\omega) = \int_{\mathcal{S}_{++}^d} \exp\left(-\frac{1}{2}\omega^\top \Sigma\omega\right) d\rho(\Sigma) \tag{21}$$

for some finite measure ρ on $\mathcal{S}_{++}^d$ (where $\Sigma = A^{-1}$).

Corollary 8.2 (Radial Case: Schoenberg-Bernstein)**.** *If $\phi(z) = \psi(\|z\|^2)$ is radial, then ϕ is representable as a mixture of* isotropic *Gaussians if and only if $\psi : [0, \infty) \to \mathbb{R}$ is completely monotone, i.e.,*

$$(-1)^n\psi^{(n)}(t) \geq 0 \quad \forall t > 0,\ \forall n \geq 0. \tag{22}$$

Theorem 8.3 (Density)**.** *The family $\{K_\theta : \theta \in \mathcal{M}_+(\mathcal{S}_+^d)\}$ is dense in the space of continuous stationary positive semi-definite kernels on $\mathbb{R}^d$, under the topology of uniform convergence on compact sets.*

9 Summary

Definition. For $\theta \in \mathcal{M}_+(\mathcal{S}_+^d)$, the integral anisotropic Gaussian kernel is:

$$K_\theta(x, y) = \int_{\mathcal{S}_+^d} \exp\left(-\frac{1}{2}(x - y)^\top A(x - y)\right) d\theta(A).$$

RKHS norm (continuous infimal convolution):

$$\|f\|^2_{\mathcal{H}_{K_\theta}} = \inf_{f=\int f_A\, d\theta(A)} \int_{\mathcal{S}_+^d} \|f_A\|^2_{\mathcal{H}_{K_A}}\, d\theta(A).$$

GP decomposition: $f \sim \mathcal{GP}(0, K_\theta) \iff f = \int f_A\, d\theta(A)$ with $f_A \overset{\text{ind}}{\sim} \mathcal{GP}(0, K_A)$.

Spectral density: $p_\theta(\omega) = \int_{\mathcal{S}_{++}^d} \frac{|A|^{1/2}}{(2\pi)^{d/2}} e^{-\frac{1}{2}\omega^\top A^{-1}\omega} d\theta(A)$.